Alloy Steel

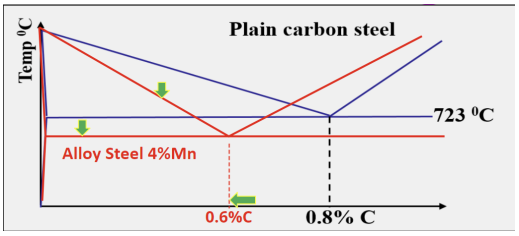
Alloy steels are plain carbon steels with other alloying elements for certain properties

A steel is considered an alloy steel if it is a plain carbon steel with

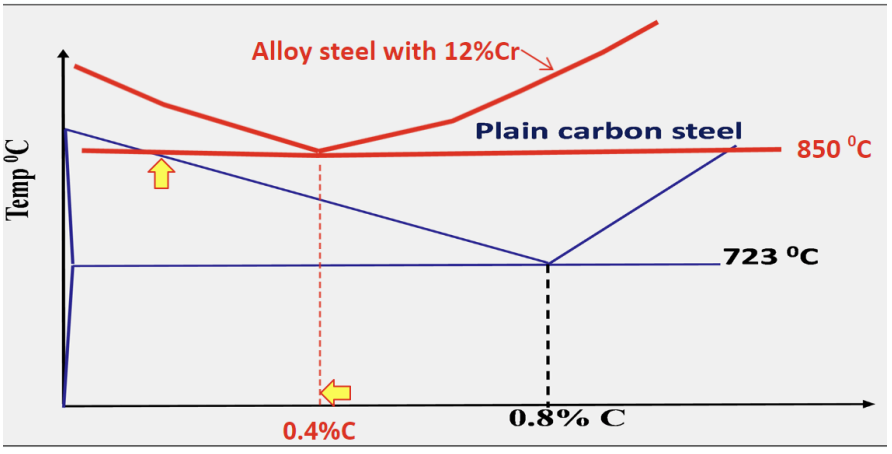
- at least 1.65% Mn
- at least 0.6% Si
- at least 0.6% Cu
- a minimum % of Cr, Mo (molybdenum), Ni is added ?? is this really necessary

Alloying Elements

- Recall carbon added to pure iron. Carbon can either be dissolved in Fe as a solid solution (ferrite) or may form compounds with iron (cementite, Fe3C)
- Similarly, different alloying elements will either form solid solutions or compounds, however with carbon instead(forming carbides)
- Elements that dissolve in ferrite are Ni, Mn, while elements that form carbides are Cr, Mo, W and V.
- **Chromium can dissolve or form carbides, however this depends on the heat treatment process used.
- Dissolving in ferrite
  - For Ni and Mn, when dissolved in ferrite, it forms a solid solution with the host iron atoms.
  - This causes an increase in strength and toughness.
  - The phase diagram of these steels also changes, with the eutectoid point shifting back (<0.8%C) and the LCT of the phase diagram being lowered



- Forming carbides
  - When Cr, Mo, W or V form carbides in the steel, the steel increases in hardness, wear resistance and strength.
  - The phase diagram also changes, with the eutectoid point moving back again however with the LCT being raised.



  - (Both decrease eutectoid point, ^ hardness, ^ LCT, maybe because strong carbides formed → harder to break ionic bonds to change phase → increased LCT???? thats kinda cool)

Steel Designation

Steels can be categorised based on an 4 digit numerical label (AISI)

For a 4 digit number ABCD,

A shows the type of steel and the major alloying elements

The type of steel for a number A is convention based:

Digit Designation	Types of steel
10xx	Plain carbon steel
23xx	Nickel steel (3.5% Ni)
31xx	Nickel chromium steel
43xx	Nickel chromium molybdenum steel

B shows the approximate percentage of the alloying elements

CD shows the % of carbon used, being 0.CD%.

Thus, for a steel AISI 2317,

It is a nickel steel(2) with ~3% nickel (3) and 0.17% carbon (17).

Note that for steels with the first digit not 1, it is an alloy steel.

Effects of Alloying Elements

Mechanical Properties

Adding Cr, V and W to form hard carbides increases strength and hardness

Adding Mn and Ni to dissolve in ferrite increases strength

Generally, adding alloying elements helps to increase strength of the steel.

Hardenability

Recall that higher carbon content $\longrightarrow$ lower CCR $\longrightarrow$ higher hardenability.

Other than higher carbon content, more alloying elements also lower CCR and increase hardenability. Generally, alloy steels have better hardenability than plain carbon steels (as more elemts present)

Thus, comparing AISI 1040 and AISI 4340, even though both have the same carbon content. as AISI 4340 is an alloy steel with more alloying elements than AISI 1040, a plain carbon steel, it has a higher hardenability

**AISI 4340 actually has the best hardenability out of all the carbon steels.

Applications of Alloy Steels

For pistons, valves, crankshafts, axles, good mix of strength and toughness is needed, thus other than medium carbon steels, Ni steel/Ni-Cr-Mo steels are used.

For (rotary) cams, roller and ball bearings, which are subject to high friction, thus high hardness and wear resistance is needed. Thus, chromium steel with 1% Cr is usually used.

Tool Steels

Tool steels refers to steels used to work and shape materials (tool steels are used in the tools themselves)

Tool steels need to have high hardness, strength, wear resistance, as well as toughness and hot hardness; the resistance to softening at higher temperature.

These steels usually contain Cr, Mo, W and V, as well as Co. These steels are hard, strong, and wear resistant, making it useful to shape other materials.

There are 4 types of tool steels used.

Shock Resisting Tool Steels

Shock resisting tool steels are used when the tool experiences repetitive and sudden impact stresses, like chisels, hammers and punches.

Toughness is the most important property for these tools, thus S type (shock) steels are used.

Cold Working Tool Steels

Cold working tool steels are used in cold working, where wear resistance is the most important.

Thus, O(oil hardened), W(water hardened) and A(air hardened) steels are used.

Hot Working Tool Steels

Hot working tool steels are used in hot working, like dies and rollers, at high temperatures. Thus, hot hardness is important to prevent the steel from softening at high temperatures.

Thus, H (hot) type steels are used.

High Speed Tool Steels

High Speed tools are used for tools that cut/shape material at high speeds, like drill bits. Thus, hardness and wear resistance is the most important. We use W(tungsten) steels and M(molybdenum).

One high speed tool steel commonly used is 18/4/1, a steel with 18%W, 4%Cr, 1%V.

Stainless Steels

Stainless steels are steels that have high corrosion resistance.

The main criteria for stainless steels is an alloy steel with at least 12%Cr and up to 26%Cr, usually with Cr being the main alloying element

The Cr forms a thin and dense Chromium oxide layer to protect the steel from corrosion.

Despite the high corrosion resistance, bleach, hydrofluoric and hydrochloric acids will still corrode stainless steels.

There are 3 main classes of stainless steels, each with different properties and structures.

Ferritic Stainless Steel

This steel has 14-26%Cr and a low carbon content of <0.2%C.

The microstructure will still be mostly ferrite, thus the properties of ferrite, being ferromagnetic, soft and ductile stay. Thus it also has good formability and machinability due to its high ductility.

Ferritic Stainless Steel is the least corrosive resistant out of all 3 stainless steels.

However, due to the carbon content (<0.2%C) is to low for hardening (at least 0.3%C), ferritic stainless steel is not heat treatable and can only be strengthened via cold working. It is also the least expensive.

This steel is mostly used for building trims, utensils(fork and spoons), deep drawn articles and kitchen sinks, aka complex and organic shaped objects (exception of building trims)

Martensitic Stainless Steel

This steel has 12-18%Cr, and a high carbon content of 0.2-1.2%C The microstructure consists of martensite.

Martensitic stainless steel can be hardened to increase strength and hardness, as well as tempered to increase toughness as well for a good combination of these properties. These steels are also magnetic.

These properties of high hardness make them useful for cutting tools like knives, cutting blades, surgical blades and scissors.

Austenitic Stainless Steel

This steel has 12-24%Cr, 8-25%Ni, however with a low carbon content. This steel's microstructure is mostly austenite, as Ni lowers the LCT, allowing there to be an austenite microstructure even at room temperature.

The most common Austenitic stainless steel is 18/8 Stainless Steel

Austenitic stainless steel is the most corrosion resistant out of all 3 of the stainless steels.

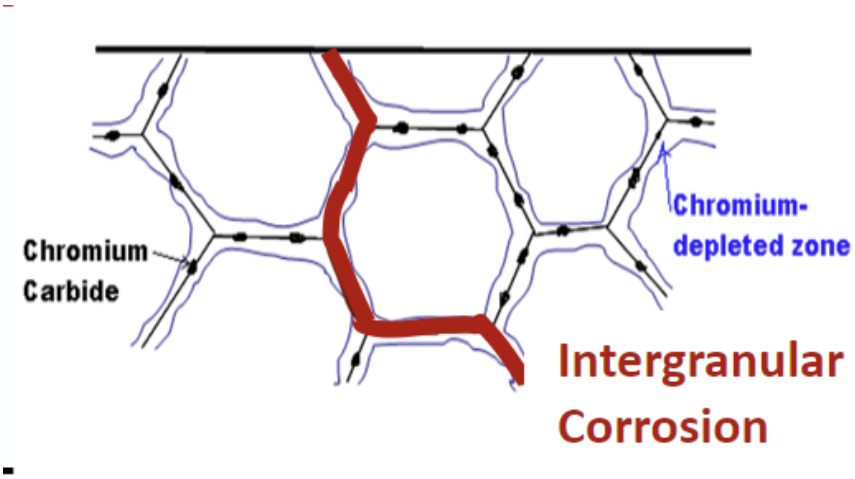
This steel are non magnetic, soft, ductile, and cannot be heat treated due to low carbon content. Thus, these steels are strengthened via cold working.

This steel are also biocompatible; aka it does not produce a toxic or injurious response in living tissue.

This steel is also susceptible to intergranular corrosion. Usually, chromium forms the chromium oxide layer to protect the steel. However, in austenitic steel, chromium carbide is likely to form along the grain boundaries. This

causes a depletion in chromium in some zones along the grain boundaries where chromium oxide has not formed, causing corrosion to occur is said boundaries.

CrCarbide instead of CrO forms along grains ⟶ less Cr present in some areas ⟶ corrosion between grains occurs.



This steel is mostly used for handling food and for surgical implantation, mainly due to the high biocompatibility.

Below summarises the properties and applications of tool and stainless steels

Steel	Shock Tool	Cold Working	Hot Working	High Speed	Ferritic Stainless	Martensitic Stainless
Composition	S type Steels	O,W or A steels	H type steels	W, M type (W and Mo elements)	14-26% Cr, <0.2%C	12-18%Cr, 0.2-1.2%C
Advantages	good toughness	good wear resistance/hardness	good hot hardness	good hardness and wear resistance	high ductility thus high malleability and formability, ferromagnetic, can be cold worked, cheapest	strong and hard, can be further hardened or tempered to manipulate properties (same properties as martensite), ferromagnetic
Disadvantages					cannot be hardened, lowest corrosion resistance	brittle
Applications	for tools undergoing repetitive impact	for tools used during cold working	for tools used during hot working	for tools that machine at high speeds, e.g. drill bits, saws	trims, utensils deep drawn articles and kitchen sinks (complex shapes/moulded by cold working)	cutting tools, knives, blades, scissors, surgical blades.
Standard Examples				18/4/1, 18%W, 4%Cr, 1%V	430	410